

FEATURE ENGINEERING

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**01 YOUR RESEARCHER -
RECAP**

**02 YOUR STRATEGY -
RECAP**

**03 FROM UNIVERSE TO
DATA**

**04 WHAT TO LOOK FOR
IN THE EXAMPLE**

03

The Feature Lab. One feature, screened on a point-in-time universe, before any book exists.

WHO READ THEM.

Hands up, work by work. Then one line from whoever read it: what did it change about your objective?

TIMELINE 01 · THE ARGUMENT

- ☐ 1970 Fama · Efficient Capital Markets: A Review of Theory and Empirical Work
- ☐ 1979 Kahneman & Tversky · Prospect Theory
- ☐ 1980 Grossman & Stiglitz · On the Impossibility of Informationally Efficient Markets
- ☐ 1997 Shleifer & Vishny · The Limits of Arbitrage
- ☐ 2004 Lo · The Adaptive Markets Hypothesis

TIMELINE 02 · THE EVIDENCE

- ☐ 2011 Khandani & Lo · What Happened to the Quants in August 2007?
- ☐ 2016 Daniel & Moskowitz · Momentum Crashes
- ☐ 2016 McLean & Pontiff · Does Academic Research Destroy Stock Return Predictability?
- ☐ 2016 Harvey, Liu & Zhu · ... and the Cross-Section of Expected Returns
- ☐ 2023 Jensen, Kelly & Pedersen · Is There a Replication Crisis in Finance?

THE RULE FROM SESSION 02 A paper you did not read is a claim you cannot cite. Mark the board; the unread ones go into your notes with read this week.

WHAT DID IT CHANGE.

Three questions, answered out loud. Write your answer down: it sharpens a claim in your `OBJECTIVE.md`.

01

WHICH CLAIM CHANGED YOUR MIND?

One paper, one claim, and the belief you held before it. If nothing moved, say which paper you would argue with.

02

WHICH ONE ARGUES AGAINST YOUR OBJECTIVE?

Crowding, crashes, decay, failed replication — four of the ten exist to attack the idea you brought. Which one lands?

03

WHAT WOULD YOU TEST FIRST?

The smallest experiment that could kill your idea. That is the feature you screen today.

THIS SESSION

UNIVERSE, DATA, ONE FEATURE.

Hands-on, start to finish: a point-in-time universe, its data on disk, one feature screened — then we stop, on purpose, before any book.

BUILD YOUR OWN RESEARCHER

Paste one line → name it → the interview → feed it the ten works

ONE PER PERSON, NOT PER STRATEGY.

Your reading is yours and outlives any one idea. Each strategy is a separate repository from the process template.

YOURS, ONE OF THEM

THE RESEARCHER

Sources/ · Knowledge/ · Philosophy/ ·
Studies/

Reads every strategy's Bibliotheca/, drafts its claims and hypotheses, and never writes into one uninvited.

STRATEGY A

MOMENTUM IN MID CAPS

Its own repo, its own universe, its own bibliography.

STRATEGY B

QUALITY AFTER EARNINGS

Same eight steps, same folder names, different idea.

STRATEGY C

THE ONE YOU WROTE LAST WEEK

Your objective from session 01 lives here.

TWO REPOS, TWO
JOBS

your researcher = what you know · your strategy = what you are testing · one package builds both

FOUR FOLDERS, ONE DIRECTION.

The arrow only points one way. That is what stops the library becoming a second opinion of itself.

Sources/

WHAT YOU READ

Books, Papers, Clippings. The researcher reads here and never writes here. PDFs stay out of git.

Knowledge/

WHAT IT READ

One note per chapter read, and a wiki page per idea. INDEX.md is read first, always; LOG.md is append-only.

Studies/

WHAT YOU WORKED OUT

An idea, a plan or a decision, from your library. Strategy work lands in the strategy itself.

Philosophy/

YOUR VOICE — THE SIDE CHANNEL

How you invest, in your own words. Cited by everything, generated from by nothing, and edited only through a diff you approve. Anything you would not want read does not go in the repository at all.

PASTE ONE LINE.

It reads the repository's setup notes and does the rest. It asks you for one thing: the name — and the name is the folder.

PASTE THIS INTO YOUR ASSISTANT

Please help me install this repo: <https://github.com/KaxaNuk/KaxaNuk-Researcher>

Behind the line: `uv tool install apm-cli==0.29.0`, then `uvx --from apm-cli==0.29.0 apm install -g KaxaNuk/KaxaNuk-Researcher --target <your assistant>`. One package, installed once for your user — it carries the researcher, every Lab skill, the strategy template and the worked example. Every folder you open afterwards has them.

TWO TOOLS FIRST

`git` · `uv`

Both on the machine before you paste. The assistant checks, and stops if one is missing.

NOTHING TO CLONE

`init-researcher` · `init-strategy` ·
`init-example`

Three skills that each make a folder. You ran the first in S00, the second in S01.

ONE CATCH

A skill is discoverable in a *new* session, never the one that installed it. Close it and reopen.

RUN THE INTERVIEW.

About three minutes, in your language. It writes `RESEARCHER.md`, and the agent and skill that make your researcher present in every session.

interview Run it again with force to start over. You will want to, once you know what you actually wanted.

01**WHO YOU ARE**

What you do, what you are here for, and the domains it should follow — coding, math, finance.

02**HOW IT SPEAKS**

Terse or thorough, in your language. And where it may push back: on evidence, never on taste.

03**WHAT YOU READ FOR**

Asked by your first read: which question a source serves. Every later source is read against them, by number.

04**THE NON-NEGOTIABLES**

What it is never allowed to do. The answer that makes it useful instead of agreeable.

THE NON-NEGOTIABLES.

Four lines from a sample RESEARCHER.md. Write your own version of these before you write anything else.

-
- 01 Every number about a strategy comes from the Lab's libraries. The researcher never computes one itself.
 - 02 A hypothesis is written before its test, and every prediction in it cites a source. No source, no prediction.
 - 03 A strategy graduates only against criteria written down beforehand. Never on a good month.
 - 04 The design of an experiment is challenged before it runs, not after it fails. Find the flaw while it is still cheap.
-

HOW IT CITES A claim with no source is written as a lead, never as a fact.

FEED IT SESSION 02.

The ten works on the two timelines, in order — each answers the one before it. Drop the PDFs into Sources/Papers/ and read them one at a time.

read plan first · your go · then it writes · contradictions flagged, never overwritten

TIMELINE 01 · THE ARGUMENT

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- 2023 Jensen, Kelly & Pedersen · Is There a Replication Crisis in Finance?

WHY THESE TEN Four of them are the reason the rest of this session is about discipline: crowding, crashes, post-publication decay and failed replication.

ONE PACKAGE, EVERY SKILL.

Every one that writes shows its plan first and waits for your go. `apm update -g` refreshes them all.

MAKE A FOLDER

`init-researcher`

Makes your home, then the interview writes RESEARCHER.md and the agent.

`init-strategy`

A new strategy from the template: eight steps as folders, empty bibliography.

`init-example`

The same scaffold, filled in — a worked strategy to study. Homework.

`update`

Brings a new version, keeps your RESEARCHER.md, Philosophy and Knowledge.

BUILD AND ASK

`read`

Sources into Knowledge, a chapter at a time — you pick which chapters. In a strategy, into its Bibliotheca.

`query <question>`

Answers from the library. Every claim cited, gaps named. Fires on its own.

`objective [strategy]`

Drafts the main idea and its claims — before any paper, then again from the notes.

`blueprint · challenge`

Thesis and predictions, before the rule; then a finished experiment checked against its own blueprint.

KEEP IT HONEST

`audit [deep]`

Read-only: links, duplicates, stale index, orphans. `deep` adds contradictions.

`refine <path>`

A voice-preserving editor pass over your own writing. Diff first.

`next`

Where you stand, and the one thing to do next.

`teach <topic>`

A multi-session tutor, grounded in what you have read.

YOUR STRATEGY

init-strategy · the order of work · the prefix law

ONE COMMAND, ONE STRATEGY.

Open your assistant in the folder that will hold your strategies and run it. It asks for the name, then builds the template around it.

`init-strategy`

The KaxaNuk Strategy Template, copied from the package on your machine: the eight steps as folders, `OBJECTIVE.md` waiting for claims, `BIBLIOGRAPHY.md` and `LOG.md` empty. No code on *main*.

01

YOURS, AND PRIVATE

Not a fork. A fresh repository you own — push it to your own GitHub, private. One folder per strategy, as many as you like.

02

THE RESEARCHER COMES ALONG

Its agent and skill are installed for your user, so it is in every session. Work lands in the strategy; nothing flows back home.

03

THE EXAMPLE IS THE SAME SHAPE

`init-example` builds a worked strategy beside yours. You read it in S01; you do not run it, and you do not copy it.

THEN, IN A NEW SESSION Open the assistant in the strategy folder, and the first command it accepts is `objective`.

OBJECTIVE FIRST.

Lettered A to H, never to be mistaken for the eight steps. C was your pre-work; today is D.

A	THE OBJECTIVE before any paper is read	objective — claims in your own words, each untested, with the question that settles it	OBJECTIVE.md	S01
B	THE READING for each claim	read, one note per paper naming the claim it serves; then objective again	Bibliotheca/	S01–S02
C	THE UNIVERSE delisted names included	the KN-TU30 seed for your sleeve, replacements logged	Universe/...csv	BEFORE
D	THE DATA curator · universe notebook · refinery · analyzer	the curator and universe.ipynb as pre-work; refinery and analyzer today; measurements into RESULTS.md	Data/	TODAY
E	THE BLUEPRINT the benchmark first, then before the rule	the benchmark entry in JOURNAL_1.md since S01; BLUEPRINT_1.md pushed before the rule	Experiments/ Experiment_1/	S04
F	THE BROAD READING what the blueprint left open	read the blueprint’s leads	BIBLIOGRAPHY.md	S04
G	THE CYCLE portfolio · backtest · attribution	the experiment notebook, until Verify passes	experiment_1.ipynb	S04–S05
H	THE RESULTS kept or rejected	every finished cycle, with the claim it moved	RESULTS.md	S06

THE PREFIX IS THE ADDRESS.

It tells you which stage owns a column — and therefore which file to open when it is wrong.

m_*	THE PROVIDER, VIA THE CURATOR Raw market data — what arrived.	Change it: never.
C_*	CURATOR CUSTOM CALCULATIONS One security's own history. Nothing else is visible.	Change it: you need a new per-security quantity.
r_*	REFINERY CUSTOM CALCULATIONS Securities against each other, per date.	Change it: you need a rank, a breadth reading, or a fitted model.
current_*	JOINED FROM THE SECURITY MASTER Today's classification — <i>not</i> point-in-time.	Change it: you group or report by something new. Never select on it.

THE TRAP Selecting on **current_*** is survivorship bias with a respectable-looking column name. It knows today's sector for a company that was something else in 2014.

FROM UNIVERSE TO DATA

the curator as pre-work · the security master · trust, then build

ONE INTERFACE, EVERY PROVIDER.

Open source, on PyPI, no account needed. It turns provider chaos into one homogenised time series per identifier.

CONFIGURED IN THE REPO

No spreadsheet. Identifiers come from `Investable_Universe.csv`, dates and columns from the strategy's own parameters — plain files, versioned with the strategy.

UNIFIED ACCESS

One interface across data providers and your own in-house files, with the tags homogenised.

PREBUILT CALCULATIONS

Returns, ratios, trend, volatility, volume — and outlier adjustments — ready to switch on.

YOUR OWN **C_*** FUNCTIONS

Custom calculations extend it. This is where your per-security feature inputs are written.

FOUR COMMANDS.

Pre-work, from the strategy folder. Environment, key, configuration, run.

01

ENVIRONMENT

```
uv sync
```

Installs the Data Curator the template declares. One command, one lock file.

Python 3.13, as everywhere in the program.

02

THE KEY

```
cp Config/.env.template  
Config/.env
```

The starter curator uses Yahoo, with no key. A key of your own, such as FMP, is optional and goes only here.

Never print a value from .env — an exposed key is rotated, not edited out.

03

THE CONFIGURATION

```
Universe/Investable_Univ  
erse.csv · Data/curator.  
py
```

The seed supplies the identifiers; the driver fixes the date window and the output columns. Ask the assistant to write them from your objective — as in the `liquid-golden-cross` example.

KN-TU30 from 2005, to 2024-12-31: the in-sample end.

04

THE RUN

```
uv run python Data/curat  
or.py
```

One structured file per identifier, aligned on one calendar, in `Data/Curator/Time_Series/`.

Nothing under `Data/` is committed. A regenerable file is not a backed-up file.

THEN, FIFTH `universe.ipynb` — profiles what the Curator downloaded and writes the security master the refinery will join. Run it, look at the coverage: the room starts from here.

STOP HERE, ON PURPOSE.

ON DISK WHEN YOU LEAVE

- Universe/ — the KN-TU30 seed and Selection_Log.csv
- custom_calculations.py — your c_* feature, with tests
- The analyzer's IC and ICIR at two horizons, summarised
- RESULTS.md — keep or drop, under “Before any experiment”
- JOURNAL_1.md — the threshold and the sign, pushed first
- TRIALS.csv — one row for every feature screened

Before you leave, every item on the left exists on your machine. Every item on the right is deliberately not done yet.

NOT TODAY

- Any book — the blueprint comes first, in session 04
- Any performance number — the engine, in session 05
- A second feature — every one screened is counted
- The holdout — 2025 onward stays sealed

A pipeline you ran once, end to end, on data you understand beats a portfolio you cannot explain. That is why we stop.

TRUST, THEN BUILD.

Six checks on what the Curator wrote, before any feature is built on it. Each one is a rule applied to every security and every date — never a judgement call on one row.

01

SPLITS AND DIVIDENDS

Adjusted prices, or one unadjusted split invents a –50% day and a momentum signal that never existed.

02

THE DEAD ARE PRESENT

Delisted names in the universe with their last price. A universe of survivors backtests beautifully and trades badly.

03

THE AVAILABILITY DATE

Fundamentals carry three dates. The one you may use is when the market could have read it — not the period end.

04

OUTLIERS, BY RULE

Statistical: is it unusual (z-score, MAD)?
Financial: is it possible (negative price, volume of zero for a year)? Both, then a rule.

05

GAPS AND STALE PRICES

A price that does not move for twenty days is not a calm stock, it is a missing feed. Coverage first, then signal.

06

EVERY CORRECTION LOGGED

Whatever you change lands in `Data_Issues.csv` with the rule that changed it. A cleaning step you cannot explain is a result you cannot defend.

WHAT TO LOOK FOR IN THE EXAMPLE

the menu of signals · the six ways a number lies to you

PICK ONE, NOT EIGHT.

Your objective already implies one of these. Build that one properly before you build a second.

RETURN SIGNALS

Earnings surprise — abnormal returns following an unexpected result, and the drift after it.

Short-term reversal — mean reversion after an extreme short-horizon move.

Residual momentum — trend persistence that survives removing factor exposure.

Quality — profitability and balance-sheet strength, priced slowly.

Accruals — mispricing tied to low earnings quality.

Sentiment — predictive content in news, filings and textual tone.

Event-driven — temporary mispricing around M&A, buybacks and guidance.

RISK FACTORS

Market — exposure to the asset class moving at all.

Beta — sensitivity of the security to that benchmark.

Sector — exposure to industry-specific economics.

Size — market capitalisation.

Value — price relative to fundamentals.

Momentum — the persistent trend, as a premium rather than an edge.

Volatility — exposure to return variability.

Growth — exposure to strong fundamental expansion.

NOTICE Momentum and quality appear on both sides. Which one yours is depends entirely on whether attribution can explain it away.

SIX WAYS IT IS A LIE.

When a result surprises you on the upside, one of these is usually why. Check them in this order.

LOOK-AHEAD

A value indexed on a date before you could have had it. The commonest, and the most flattering.

SURVIVORSHIP

Your universe is who is still listed. The delisted names are the ones the strategy would have bought.

SELECTION ON **CURRENT_***

Filtering history by today's classification. Survivorship with a respectable column name.

UNDECLARED SWEEPS

Forty lookbacks tried, one reported. The p-value you quote is for the one test you pretended to run.

UNCOSTED TURNOVER

A signal that rebalances daily on illiquid names is a spread-payment schedule, not a strategy.

BORROWED BETA

It works, and it is sector beta. Real, and not a return of your own. Attribution settles this in session 05.

ASK YOUR RESEARCHER /query which of these six does my feature risk, and what did we read about it?

UP NEXT

NEXT: PORTFOLIO CONSTRUCTION.

Session 04 pushes your blueprint before the rule, then builds an equal-weight book that passes its invariants. It starts from your `RESULTS.md`.

TO DO.

Three deliverables and one thing not to do. Session 04 opens with your blueprint draft.

01 FINISH THE SCREEN

Keep or drop, written down.

- The measurements in RESULTS.md, under “Before any experiment”
- Every feature you screened, a TRIALS.csv row
- Which number the seeded leak inflated, in JOURNAL_1.md

02 READ THE EXAMPLE'S BLUEPRINT

Read it; do not run it.

- `init-example`, beside your strategy, as in S01
- Each prediction, and the note or measurement it cites
- Its control: what differs from the rule, in one thing

03 DRAFT YOUR BLUEPRINT

`/blueprint 1`, drafted, not pushed.

- It needs only your RESULTS.md and your notes
- Every prediction cites a note or an analyzer measurement
- A control that differs from the rule in one thing
- In S04 you answer the critic, then push it before the rule

04 DO NOT ACTIVATE THE DEMO

The Lab demo waits for session 04.

- Requested in S00, not activated
- Its 14 days start in S04 and must cover S04 to S06

WHERE IT GOES Results, journal, trials and the blueprint draft into your strategy repository. The researcher stays on your own machine — your library is private to you.

BACK TO THE PILLARS.

The coding, markets and math sessions behind today. Go back to one when you could not do its part with the AI off.

CODING · CODING FOUNDATIONS

- **CF S05 NumPy, pandas and Notebooks**
Jupyter, modules, stripped outputs, DatetimeIndex
- **CF S06 Time Without Look-Ahead**
shift, warm-up nulls, `index < date`, per-date ranks
- **CF S07 Features as Data Curator Functions**
`c_*` features with `DataColumn`
- **CF S03 Functions, Types and Tests**
tests as tables, the platform door
- **CF S12 Capstone: Ship a Tested Release**
your tested `c_*` candidate

MARKETS · FINANCIAL MARKETS

- **FM S06 Indices, Benchmarks, ETFs and Survivorship**
survivorship, point-in-time membership
- **FM S01 The Ecosystem and the Security Master**
the security master
- **FM S02 Prices, Total Returns and Time Value**
adjusted and unadjusted prices
- **FM S05 Statements, Valuation and the Three Dates**
only for a fundamentals feature
- **FM S07 Liquidity, Trading Costs and Capacity**
the traded-value proxy
- **FM S09 Risk, Diversification and the Long-Only Book**
correlation, for the diversification check
- **FM S11 Factor Models: Risk Factors and Return Signals**
classifying your candidate feature

MATH · MATHEMATICAL FINANCE

- **MF S08 Cross-Sectional Signals: IC, Breadth and Fama-MacBeth**
MAD z-scores, ranks, IC, ICIR, the fundamental law
- **MF S05 Linear Algebra and Covariance**
correlation, for the diversification check
- **MF S07 Time Series and Trend Signals**
look-ahead as causality
- **MF S11 Supervised Learning Without Leakage**
a machine-learning feature, without leakage

THE READING LIST.

Read two. Compile them. If you only read one, make it Arnott, Harvey & Markowitz — it is a checklist disguised as a paper.

THE DISCIPLINE · WHY A FEATURE MUST BE DECLARED BEFORE IT IS TESTED

ARNOTT, HARVEY & MARKOWITZ

2019 A Backtesting Protocol in the Era of Machine Learning

HARVEY, LIU & ZHU

2016 ... and the Cross-Section of Expected Returns

MCLEAN & PONTIFF

2016 Does Academic Research Destroy Stock Return Predictability?

THE CRAFT · HOW THE COLUMNS ARE ACTUALLY BUILT

PALEOLOGO

2025 The Elements of Quantitative Investing
2021 Advanced Portfolio Management

GRINOLD & KAHN · JANSEN

2000 Active Portfolio Management, 2nd ed.: Information Analysis
2020 Machine Learning for Algorithmic Trading, 2nd ed.: Financial Feature Engineering

KAXANUK · VIAFORE

2026 Data Curator, README
2021 Robust Python, for the custom calculations you will rely on

NOW YOU HAVE SOMEWHERE TO PUT THEM Sources/Papers/ → read → Knowledge/

TWO FOLDERS. ONE PACKAGE.

What you know, and what you are testing. A feature that cannot name its source is not finished.

```
github.com/KaxaNuk/KaxaNuk-Researcher
```

Paste the URL, name it. One per person.

```
init-strategy · init-example
```

From the same package. One per strategy.

QUESTIONS research@kaxanuk.mx

NEXT

SESSION 04

TOPIC

**PORTFOLIO
CONSTRUCTION:
BLUEPRINT BEFORE
THE RULE**

YOU LEAVE WITH

**BLUEPRINT_1.MD
PUSHED FIRST, AND AN
EQUAL-WEIGHT BOOK
THAT PASSES ITS
INVARIANTS.**

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